Module 5: Regularized Regression

Part 2: Lasso Regression

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Why does ridge regression improve over least squares?

Ridge regression's advantage over least square is rooted in the **bias-variance trade-off**.

$$\hat{\beta}^R = \min_{\beta}\left(\sum_{i=1}^{n}\left(Y_i - (\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)\right)^2 + \lambda \sum_{j=1}^{p} {\beta_j}^2\right), \lambda > 0$$

- As λ increases, the flexibility of the ridge regression fit decreases, leading to a decreased variance but increased bias.

↳ each predictor pays a price to stay in the model

↳ ℓ_2 penalty → if the predictor can't afford the price, the model compensates by shrinking the regression coef. which protects from overfitting

Ridge regression recap

- Minimizes RSS + l_2 penalty term.
- Shrink coefficients towards 0 (but does not set them exactly to zero).
- The amount of shrinkage is controlled by λ.
- Adds bias, but reduces variance.
- Improves prediction performance.
- Handles multicollinearity well.
- Ridge regression performs particularly well when there is a subset of true regression coefficients that are mostly **small.**

Disadvantage of ridge regression

↳ shrink $\hat{\beta}_j$ towards 0, but it can not set them to 0 exactly

The Lasso

- Resolve disadvantage of ridge regression
- Performs both $\underbrace{\text{model selection}}_{\text{set } \beta_j \text{ to zero}}$ and regularization.

The Lasso

$$\hat{\beta}^L = \min_{\beta}\left(\overbrace{\sum_{i=1}^{n}\left(Y_i - (\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)\right)^2}^{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|\right), \lambda > 0$$

- $\lambda = 0$: same as least squares (min. RSS)
- $\lambda \to \infty$: all coefficeients $\hat{\beta}^L = 0$ [shrink ($\to 0$)]
- Lasso can remove predictors from the model.

$p \to d$ $(d < p)$: spasity (more simple, less predictors)

↳ more simple

The Lasso

For a λ in between the extreme, we are balancing two ideas:

- Fitting a linear model of Y on X. → min. RSS
- Shrinking the coefficients (l_1 penalty can shrink some to 0).

Lasso has no analytical solution (no closed-form formula).

- Can find lasso regression coefficients using numerical algorithms.

Lasso regression coefficients

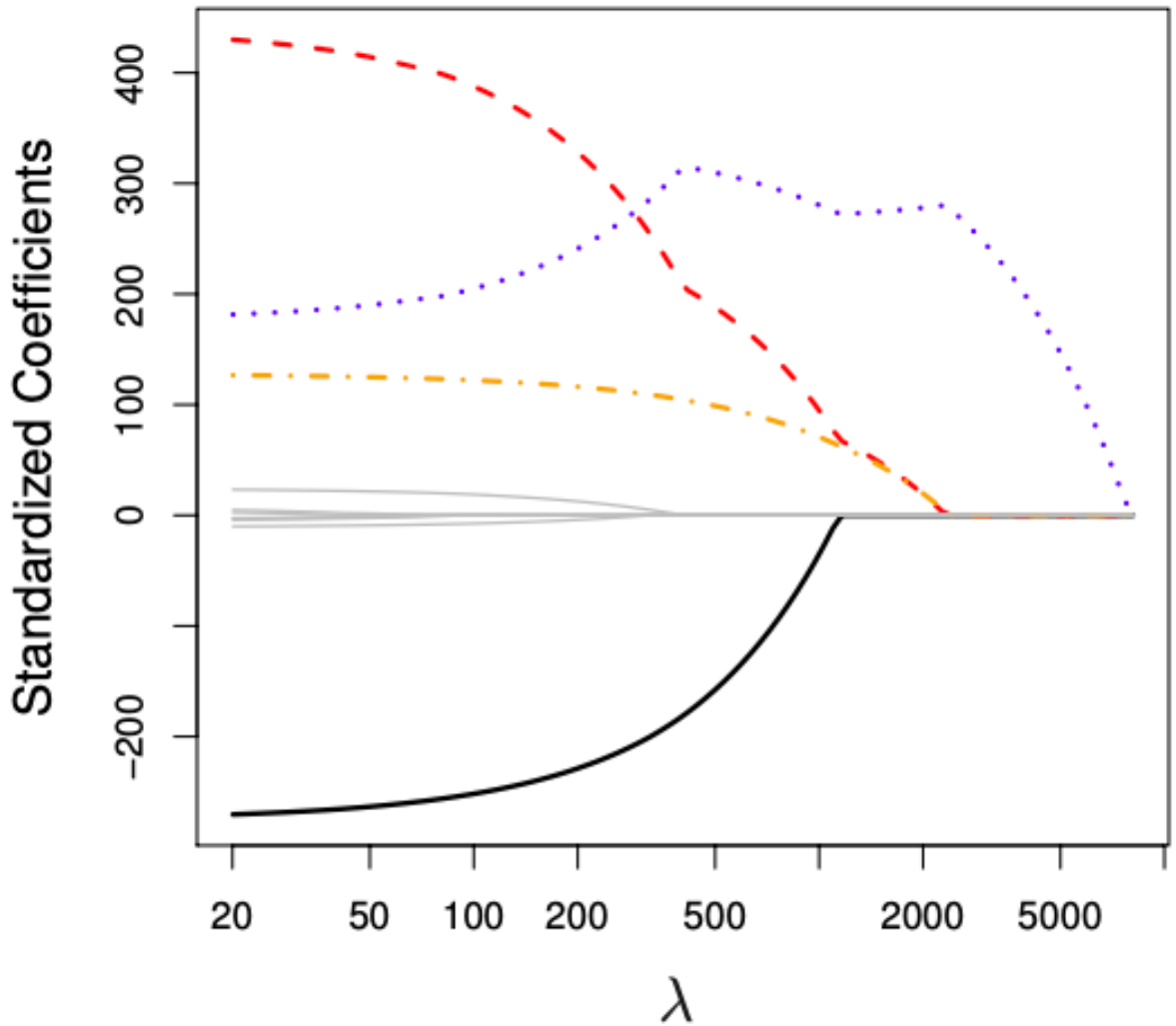

Figure 1. *The standardized lasso coefficients on the* Credit *data set are shown as a function of* λ

Lasso vs. Ridge

- In terms of prediction error (test MSE), the lasso performs comparably to ridge regression.
- Lasso penalty can set some coefficients to 0 when λ is sufficiently large.
  - Performs automatic model selection.
  - Leads to sparse models.
- Selecting λ here is (again) critical and can be done using **cross-validation.**
- Lasso implicitly assumes that a number of the coefficients truly equal zero.

Lasso vs. Ridge

- Neither method universally dominates the other
- **Lasso** performs better when only a **small number of predictors** have large effects
- **Ridge** performs better when **many predictors** contribute to the response
- In practice, the true model structure is **unknown**
- Choice of method should be guided by **prediction performance**
- Use **cross-validation** to select the better approach

See R script: `regularized_regression_code.R`

Predictive modeling tools at your disposal

(1) Least squares linear regression

- Analytical solution, simple to implement.
- Can extend to model non-linear relationships (polynomial regression)
- Model selection (subset selection, forward, backward, stepwise, cross-validation)
- Inference is straightforward to carry out.

However, high variance → overfitting

Predictive modeling tools at your disposal

(2) Ridge Regression

- Useful in improving prediction accuracy.
- Will always result in **a full model with all p predictors**. Ridge regression is the obvious choice if you believe all predictors are somewhat important.
- Can handle multicollinearity.
- Inference can also be done (relatively straightforwardly).

Predictive modeling tools at your disposal

(3) The Lasso → works best when only a few predictors are important

- Regularizes and performs model selection.
- Generally works well when only a subset of predictors are actually important.
- Inference not as straightforward to carry out.

No one approach will universally dominate the other.

Extension of Lasso: group Lasso

- Regular Lasso: selects variables one by one
- **Group Lasso: selects variables in** groups

Examples

- dummy variable for categorical variable
- Genes are grouped into biological pathways (e.g., immune, metabolism)

Group Lasso will:

- keep the group
- drop the group

Extension of Lasso: Elastic Net

$$\hat{\beta}^R$$

$$= \min_{\beta} \left(\sum_{i=1}^{n} \left(Y_i - \left(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \right) \right)^2 + \lambda \left(\frac{1-\alpha}{2} \cdot \sum_{j=1}^{p} \hat{\beta}_j^2 + \alpha \cdot \sum_{j=1}^{p} |\beta_j| \right) \right), \lambda > 0$$